



Literature Review

Sanjay M

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Paper 1

Title: A Zero-Trust Access Control Model Based on Attribute and Dynamic Trust Evaluation for Cloud Environments

Authors: Yihuan Mao, Wei Fu, Yue Zhao, Zhimin Yuan, Zhihong Sun, Yang Zhao

Publication: Symmetry, Volume 17, Issue 12, Article 2059

Publisher: MDPI

Paper Link: <https://www.mdpi.com/2073-8994/17/12/2059>

Published: December 2025

Review: In this paper, the authors addressed the problem of providing flexible and dynamic access control in cloud environments where traditional Attribute-Based Access Control can lack sufficient adaptability to changing security conditions. They proposed a Zero-Trust Access Control model based on attributes and dynamic user trust scores, called AT-ZTAC. The approach combines attribute-based access control with continuous trust evaluation so that access decisions can consider both user attributes and the current trustworthiness of the requesting entity. The authors also considered security risks associated with the Policy Decision Point, which is an important component of Zero Trust access control. The proposed model dynamically evaluates users and adjusts access decisions according to changes in their behaviour and security context. The study evaluates the proposed approach through experiments designed to examine its access-control effectiveness and security characteristics. The results show that combining attribute information with dynamic trust evaluation can provide more adaptive access decisions than relying on static access rules alone. Important challenges include maintaining accurate trust scores, protecting the policy decision process, handling changing user behaviour, and avoiding incorrect authorization decisions. Therefore, this paper is highly relevant to the ZeroTrustCloud project because it directly supports identity-centric access control, continuous trust evaluation, dynamic authorization, least-privilege enforcement, and policy-based security in cloud environments.

Paper 2

Title: A federated learning-based zero trust intrusion detection system for Internet of Things

Authors: Danish Javeed, Muhammad Shahid Saeed, Muhammad Adil, Prabhat Kumar, Alireza Jolfaei

Publication: Ad Hoc Networks, Volume 162, Article 103540

Publisher: Elsevier

Paper Link: <https://www.sciencedirect.com/science/article/pii/S1570870524001513>

Published: September 2024

Review:

In this paper, the authors addressed the problem of protecting large-scale Internet of Things environments where many distributed devices generate security data and where centralized intrusion detection can create privacy and scalability problems. They proposed a federated-learning-based Zero Trust intrusion detection system that allows different IoT environments to collaboratively train security models without directly sharing their local data. The approach combines Federated Learning with Zero Trust principles to continuously evaluate participating devices and detect suspicious behaviour. The research focuses on extracting both spatial and temporal characteristics from IoT security data so that the system can recognize different types of cyber threats. The authors evaluated the proposed approach using cybersecurity datasets and compared it with alternative intrusion-detection methods. The study shows that combining federated learning with Zero Trust can improve collaborative intrusion detection while reducing the need to centralize sensitive data. The approach is particularly useful for distributed IoT environments where devices and network conditions can change frequently. Important challenges include heterogeneous datasets, communication overhead between participating clients, changing attack patterns, and maintaining reliable trust evaluation across distributed devices. Therefore, this paper is relevant to the ZeroTrustCloud project because it demonstrates how machine learning can support continuous monitoring, trust evaluation, distributed security analysis, and automated security decisions in a Zero Trust environment.

Paper 3

Title: Asynchronous federated learning based zero trust architecture for the next generation industrial control systems

Authors: Fei Lv, Hangyu Wang, Zhiwen Pan, Rongkang Sun, Shuaizong Si, Weidong Zhang, Shichao Lv, Limin Sun

Publication: Computer Networks, Volume 269, Article 111459

Publisher: Elsevier

Paper Link: <https://www.sciencedirect.com/science/article/pii/S1389128625004268>

Published: September 2025

Review:

In this paper, the authors addressed the difficulty of applying Zero Trust security to next-generation Industrial Control Systems, where devices are heterogeneous, network structures are dynamic, and security decisions often have strict real-time requirements. They proposed FedZTA, a Zero Trust Architecture based on asynchronous federated deep learning. The framework evaluates trust using multiple types of information, including subject attributes, object attributes, cyber information from network traffic, and physical information obtained from sensors and control devices. A deep-learning model combining Convolutional Neural Network and Multilayer Perceptron techniques is used for trust evaluation. To reduce the waiting time normally associated with federated-learning aggregation, the authors proposed a grouping-based asynchronous federated-learning algorithm that groups agents according to computing capability and response requirements. The approach was evaluated using real industrial datasets from a laboratory gas-pipeline environment together with a public dataset. The results showed improved trust-decision accuracy and reduced average aggregation latency compared with the evaluated alternative approaches. The study demonstrates that physical and cyber information can be combined to improve continuous trust evaluation. Important challenges include heterogeneous industrial devices, strict latency requirements, dynamic environments, and maintaining reliable trust decisions while data remains distributed. Therefore, this paper is relevant to ZeroTrustCloud because it demonstrates continuous trust evaluation, machine-learning-based policy decisions, distributed security analysis, and adaptive Zero Trust enforcement.

Paper 4

Title: Federated Learning and Zero-Trust Architecture for Privacy Preservation and Collaborative Threat Intelligence

Authors: Alireza Chamkoori, AbdolRasool Ghasemi, AbdolRasool Ghasemi

Publication: Journal of Computer Networks and Communications, Volume 2026, Article 8893656

Publisher: Wiley

Paper Link: <https://onlinelibrary.wiley.com/doi/10.1155/jcnc/8893656>

Published: July 2026

Review:

In this paper, the authors addressed the problem of securely sharing cyber threat intelligence between distributed organizations without requiring them to exchange sensitive raw security data. They proposed an integrated Federated Learning and Zero Trust Architecture framework called FL-ZTA. The framework combines decentralized federated learning, continuous Zero Trust verification, trust-aware aggregation, adaptive policy enforcement, and secure coordination between participating organizations. The authors considered threats such as malicious clients, model poisoning, inference attacks, unauthorized participation, and communication attacks. The experimental evaluation used the CIC-IDS2017 cybersecurity dataset and considered both IID and non-IID federated-learning environments. The study compared federated-learning approaches including FedAvg, FedProx, and differentially private federated averaging within the proposed Zero Trust framework. The results demonstrated improved resistance to poisoning and inference attacks while maintaining collaborative learning performance without requiring organizations to share their original security data. The approach also provides a mechanism for continuously verifying participating clients rather than trusting them permanently. Important challenges include heterogeneous data distributions, malicious participants, model-update privacy, trust management, and maintaining good performance across distributed organizations. Therefore, this paper is relevant to ZeroTrustCloud because it supports continuous verification, distributed threat intelligence, privacy-preserving security analysis, adaptive policy enforcement, and trust-aware cloud security.

Paper 5

Title: A Zero Trust Driven Federative Learning Algorithm for Privacy Enhancement

Authors: Beverly Pule, Bakhe Nleya, Khulekani Sibiya

Publication: Applied Sciences, Volume 16, Issue 8, Article 3872

Publisher: MDPI

Paper Link: <https://www.mdpi.com/2076-3417/16/8/3872>

Published: April 2026

Review:

In this paper, the authors addressed privacy and security risks in federated-learning environments where distributed clients may be exposed to membership inference attacks, property inference attacks, and other attempts to extract information from model updates. They proposed a Zero-Trust Driven Federated Learning Algorithm for Privacy Enhancement, called ZT-FL-PE. The framework combines Zero Trust authentication and continuous verification with federated learning so that participating clients are not automatically trusted throughout the learning process. The approach calculates dynamic trust information based on client behaviour and uses trust-aware aggregation and privacy-preserving update transformations. The researchers evaluated the method using multiple datasets and compared it with different privacy-preserving approaches. The results showed that the proposed approach could improve privacy protection while maintaining model accuracy and avoiding the large computational cost associated with some differential-privacy techniques. The authors reported approximately a 5% increase in total client runtime compared with an undefended baseline, while server-side trust aggregation remained manageable. The paper also provides a convergence analysis showing how the security and privacy mechanisms affect model optimization. Important challenges include balancing privacy with model accuracy, managing trust scores, handling heterogeneous clients, and controlling verification overhead. Therefore, this paper is relevant to ZeroTrustCloud because it demonstrates how Zero Trust principles can be integrated with machine learning to provide continuous verification, adaptive trust evaluation, privacy protection, and secure distributed computation.

Paper 6

Title: Microsegmented Cloud Network Architecture Using Open-Source Tools for a Zero Trust Foundation

Authors: Sunil Arora, John Hastings

Publication: 2024 17th International Conference on Security of Information and Networks

Publisher: IEEE

Paper Link: <https://doi.org/10.1109/SIN63213.2024.10871361>

Published: December 2024

Review:

In this paper, the authors addressed the problem of securing modern multi-cloud environments where applications, containers, virtual machines, and cloud services are distributed across different infrastructure locations. They proposed a cloud-network architecture based on Zero Trust principles and micro-segmentation. The architecture provides authentication, authorization, encrypted communication, and controlled connectivity between workloads while reducing unnecessary communication paths. The authors considered multiple cloud deployment models and different types of computing resources including containers, virtual machines, IaaS, PaaS, and cloud-native services. Open-source networking and security tools were considered to avoid dependence on a single vendor and to provide flexibility when implementing the proposed architecture. Micro-segmentation was used to divide the cloud environment into smaller security zones so that compromise of one workload does not automatically provide access to other resources. The architecture focuses on controlling communication between workloads and applying Zero Trust security principles across distributed environments. The paper demonstrates how segmentation can strengthen cloud security and reduce lateral movement opportunities. Important challenges include managing policies across multiple cloud environments, maintaining secure communication, handling distributed workloads, and controlling the operational complexity introduced by micro-segmentation. Therefore, this paper is relevant to ZeroTrustCloud because it directly supports micro-segmentation, multi-cloud security, workload isolation, identity-based access control, encrypted communication, and Zero Trust network design.

Paper 7

Title: Flexible zero trust architecture for the cybersecurity of industrial IoT infrastructures

Authors: Claudio Zanasi, Silvio Russo, Michele Colajanni

Publication: Ad Hoc Networks, Volume 156, Article 103414

Publisher: Elsevier

Paper Link: <https://www.sciencedirect.com/science/article/pii/S1570870524000258>

Published: April 2024

Review:

In this paper, the authors addressed the problem of securing heterogeneous Industrial IoT infrastructures where devices have different capabilities, strict real-time requirements, and distributed operational environments. They proposed a flexible Zero Trust architecture based on network micro-segmentation, Software-Defined Networking, and centralized security management. The SDN layer provides a common abstraction for enforcing security policies across different environments, while the security-management layer controls the execution of policies and manages the network configuration. The authors implemented a prototype to evaluate the feasibility of the proposed architecture and studied how the system behaves when individual components fail. The proposed peer-to-peer SDN architecture uses asynchronous policy distribution so that security policies can continue to operate even when some components become unavailable. The experimental results demonstrated resilience to individual failures and supported decentralized operation while maintaining centralized management of network topology and security policies. Important challenges include heterogeneous IIoT devices, real-time requirements, distributed decision-making, policy management, and maintaining availability during component failures. Therefore, this paper is relevant to ZeroTrustCloud because it provides practical ideas for micro-segmentation, policy enforcement, distributed cloud environments, continuous security control, and resilient Zero Trust architecture.

Paper 8

Title: SECURING APIS IN CLOUD-NATIVE APPLICATIONS WITH ZERO TRUST PRINCIPLES

Authors: Sujeeth Reddy Pasham

Publication: International Journal of Computer Engineering and Technology, Volume 16, Issue 1, Pages 3592–3608

Publisher: IAEME Publication

Paper Link: https://iaeme.com/Home/article_id/IJCET_16_01_248

Published: February 2025

Review:

In this paper, the author addressed the problem of securing APIs in cloud-native applications where distributed microservices, containerized workloads, and multi-cloud deployments create a large and dynamic attack surface. The paper examines how Zero Trust principles can be applied directly to API security instead of relying mainly on network location or perimeter controls. The approach discusses authentication and authorization technologies including mutual TLS and OAuth 2.0, together with API-level encryption, runtime anomaly detection, continuous security testing, and monitoring. The author also examines the role of service meshes in Kubernetes environments for controlling service-to-service communication and enforcing security policies. The paper considers cloud-native database systems and multi-cloud API-management requirements as part of the security architecture. The proposed approach emphasizes continuous verification of API requests and stronger identity-based authorization between services. It also connects API security with DevSecOps practices so that security checks can be integrated throughout application development and deployment. The study identifies challenges including managing APIs across multiple clouds, maintaining consistent policies, protecting service identities, detecting abnormal API behaviour, and integrating security controls without affecting application performance. Therefore, this paper is relevant to ZeroTrustCloud because APIs and microservices are important parts of cloud-native environments and the paper directly supports identity-centric authorization, mTLS, service-mesh security, continuous monitoring, and Zero Trust policy enforcement.

Paper 9

Title: Enhancing cloud-native DevSecOps: A Zero Trust approach for the financial sector

Authors: Daemin Shin, Jiyeon Kim, I Wayan Adi Juliawan Pawana, Ilsun You

Publication: Computer Standards & Interfaces, Volume 93, Article 103975

Publisher: Elsevier

Paper Link: <https://www.sciencedirect.com/science/article/abs/pii/S0920548925000042>

Published: April 2025

Review:

In this paper, the authors addressed the challenge of applying Zero Trust security throughout cloud-native DevSecOps environments used by financial organizations. They studied the relationship between microservice-based DevSecOps environments and Zero Trust security principles and examined how Zero Trust can be applied throughout the Software Development Life Cycle. The researchers analysed eight SDLC stages and identified security policies and technologies that can support Zero Trust at each stage. Based on this analysis, they proposed a framework consisting of seven security domains and 23 subcategories covering security policies and technologies for cloud-native DevSecOps environments. The framework considers security throughout development, testing, deployment, and operation rather than treating Zero Trust only as a network-security mechanism. The authors evaluated the appropriateness of the proposed framework against the security requirements identified in their analysis. The study highlights the importance of integrating identity, access control, monitoring, policy enforcement, and security automation throughout the development lifecycle. Important challenges include the complexity of applying Zero Trust consistently across SDLC stages, managing multiple security technologies, maintaining policy consistency, and adapting controls to financial-sector requirements. Therefore, this paper is relevant to ZeroTrustCloud because it connects Zero Trust with cloud-native application development, DevSecOps, microservices, continuous verification, policy enforcement, and secure deployment processes.

Paper 10

Title: The significance of artificial intelligence in zero trust technologies: a comprehensive review

Authors: Deepa Ajish

Publication: Journal of Electrical Systems and Information Technology, Volume 11, Article 30

Publisher: Springer Nature

Paper Link: <https://link.springer.com/article/10.1186/s43067-024-00155-z>

Published: August 2024

Review:

In this paper, the author examined how Artificial Intelligence can support Zero Trust security in modern cloud and distributed environments. The paper addresses the limitations of traditional perimeter-based security and explains how Zero Trust requires continuous verification, dynamic authentication, least-privilege access, monitoring, and context-aware security decisions. The review investigates the use of AI in areas such as user and device authentication, anomaly detection, behavioural analysis, risk assessment, Zero Trust Network Access, and automated security responses. AI can process large amounts of security information collected from users, devices, workloads, and network activity and can help identify behaviour that differs from established patterns. The paper also discusses how AI-based anomaly detection can trigger additional verification when suspicious activity is detected and can support automated responses to security events. The author considers Zero Trust across cloud, hybrid, and distributed environments and highlights the importance of visibility, analysis, automation, orchestration, and governance. The review identifies challenges including the quality of security data, false positives, model reliability, privacy, explainability, adversarial attacks against AI systems, and the need for continuous monitoring. Therefore, this paper is relevant to ZeroTrustCloud because it provides a strong connection between AI-based behavioural analysis and Zero Trust, supporting the project's planned continuous monitoring, identity-centric decisions, risk evaluation, automated policy enforcement, and adaptive security capabilities.